

Horizon: Privacy-Preserving On-Device Child Safety Risk Detection via Knowledge Distillation

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2026-05-12

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Abstract

We present **Horizon**, a dual-model system for real-time child safety risk detection in messaging environments. The system comprises a high-capacity server model (Llama 3.2 3B fine-tuned with QLoRA) and a lightweight on-device classifier (MobileBERT, ~25MB) trained via knowledge distillation. Horizon detects seven risk categories—grooming, bullying, sexual content, isolation, personal information disclosure, platform migration, and threats—with severity grading and token-level explainability. The mobile variant achieves sub-100ms inference on commodity smartphones while keeping all message content on-device, eliminating the need to transmit sensitive conversations to external servers. We detail the training methodology, distillation process, and deployment architecture.

1 Introduction

Online child safety remains a critical challenge. Existing solutions broadly fall into two categories: server-side content moderation (which requires transmitting private messages to third parties) and keyword-based filtering (which suffers from high false-positive rates and trivial circumvention). Horizon addresses both limitations by deploying a neural classifier directly on the user’s device.

Our contributions are:

1. A fine-tuning methodology for adapting large language models to structured risk classification tasks
2. A knowledge distillation pipeline that compresses a 3B-parameter teacher into a 24M-parameter student with minimal accuracy loss
3. Token-level explainability via Integrated Gradients, enabling transparent risk attribution
4. Cross-platform deployment via ONNX Runtime Mobile for iOS and Android

2 Problem Formulation

Let $\mathcal{C} = \{c_1, c_2, \dots, c_n\}$ denote a conversation consisting of n messages. We define the risk detection task as learning a function:

$$f : \mathcal{C} \rightarrow (\hat{y}, \hat{s}, \hat{p})$$

where:

- $\hat{y} \in \{1, \dots, K\}$ is the predicted risk category ($K = 8$, including “benign”)
- $\hat{s} \in \{1, \dots, S\}$ is the predicted severity level ($S = 5$: none, low, medium, high, critical)
- $\hat{p} \in [0, 1]$ is the confidence score

2.1 Risk Taxonomy

We define seven non-benign risk categories based on established child safety frameworks:

Category	Definition
Grooming	Incremental boundary violations, trust building, secrecy demands
Bullying	Repeated intimidation, social exclusion, humiliation
Sexual Content	Explicit or suggestive material directed at minors
Isolation	Attempts to separate a child from support networks
Personal Info	Solicitation of identifying information (address, school, photos)
Platform Migration	Requests to move communication to less-monitored channels

Category	Definition
Threats	Direct or implied threats of physical or psychological harm

3 Model Architecture

3.1 Teacher Model: Horizon-Full

The teacher model is based on Llama 3.2 3B-Instruct, fine-tuned using Quantization-aware Low-Rank Adaptation (QLoRA).

3.1.1 Low-Rank Adaptation

Given a pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA decomposes the update as:

$$W = W_0 + \Delta W = W_0 + BA$$

where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$ is the rank. The forward pass becomes:

$$h = W_0 x + \frac{\alpha}{r} B A x$$

where α is a scaling hyperparameter. We apply LoRA to all linear projections in the attention and MLP layers:

$$\mathcal{T} = \{W_Q, W_K, W_V, W_O, W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}\}$$

Configuration:

Parameter	Value
Rank (r)	128
Alpha (α)	256
Dropout	0.05
Target modules	$ \mathcal{T} = 7$ per layer
Trainable parameters	194.5M (5.7% of total)

3.1.2 Training Configuration

Training uses the AdamW optimizer with fused CUDA kernels:

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} - \lambda \theta_t$$

where $\hat{m}_t$ and $\hat{v}_t$ are bias-corrected first and second moment estimates, $\eta = 2 \times 10^{-4}$ is the learning rate with cosine decay, and $\lambda = 0.01$ is the weight decay coefficient.

Training schedule:

- Total steps: 15,000
- Effective batch size: 32 (micro-batch 4×8 gradient accumulation steps)
- Sequence length: 2,048 tokens
- Precision: bfloat16 with TF32 tensor cores
- Gradient checkpointing enabled

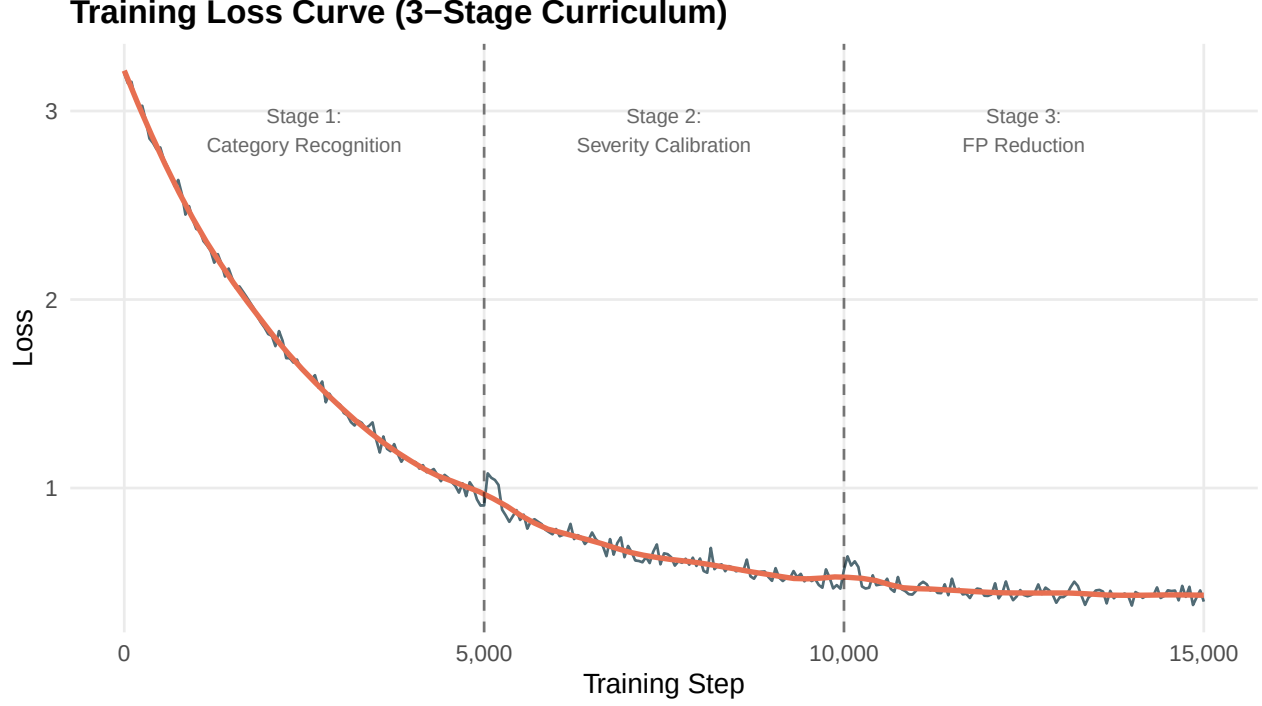


Figure 1: Expected training loss curve over 15,000 steps with three-stage curriculum learning. Vertical dashed lines indicate stage transitions.

3.2 Student Model: Horizon-Mobile

The student model uses MobileBERT as its encoder backbone with two classification heads sharing the pooled [CLS] representation.

3.2.1 Architecture

$$\text{HorizonMobile}(x) = \begin{pmatrix} \text{softmax}(W_c \cdot h_{[\text{CLS}]} + b_c) \\ \text{softmax}(W_s \cdot h_{[\text{CLS}]} + b_s) \end{pmatrix}$$

where:

- $h_{[\text{CLS}]} \in \mathbb{R}^{512}$ is MobileBERT’s pooled output
- $W_c \in \mathbb{R}^{8 \times 512}$, $b_c \in \mathbb{R}^8$ (category head)
- $W_s \in \mathbb{R}^{5 \times 512}$, $b_s \in \mathbb{R}^5$ (severity head)

Model statistics:

Property	Value
Parameters	24.6M
ONNX size (INT8)	~25MB
Max sequence length	256 tokens
Inference latency	<100ms (mobile)

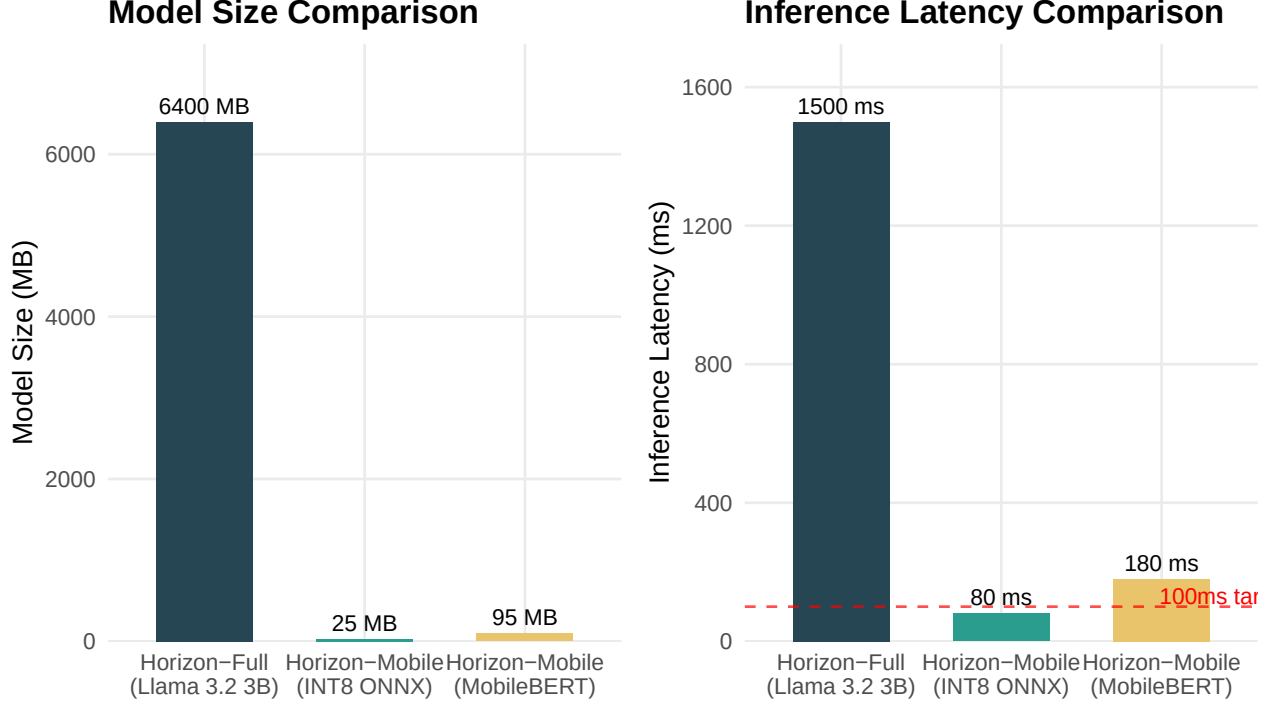


Figure 2: Model size and latency comparison between Horizon variants.

4 Knowledge Distillation

We transfer knowledge from the teacher (Horizon-Full) to the student (Horizon-Mobile) using a combination of soft label matching and hard label supervision.

4.1 Soft Label Generation

Given a training example x_i , the teacher produces a probability distribution over categories:

$$q_i = \text{softmax}\left(\frac{z_i^T}{\tau}\right)$$

where z_i^T are the teacher’s logits and τ is the temperature parameter. Higher temperatures produce softer distributions that convey more information about inter-class similarities.

4.2 Distillation Loss

The student is trained to minimize a weighted combination of two objectives:

$$\mathcal{L} = \alpha \cdot \mathcal{L}_{\text{KD}} + (1 - \alpha) \cdot \mathcal{L}_{\text{CE}}$$

4.2.1 KL Divergence (Soft Labels)

$$\mathcal{L}_{\text{KD}} = \tau^2 \cdot D_{\text{KL}}(q_i \parallel p_i^\tau) = \tau^2 \sum_{j=1}^K q_{ij} \log \frac{q_{ij}}{p_{ij}^\tau}$$

where $p_i^\tau = \text{softmax}(z_i^S/\tau)$ is the student’s temperature-scaled output. The τ^2 factor compensates for the reduced gradient magnitude at higher temperatures.

4.2.2 Cross-Entropy (Hard Labels)

$$\mathcal{L}_{\text{CE}} = \frac{1}{2} [\mathcal{L}_{\text{CE}}^{\text{cat}} + \mathcal{L}_{\text{CE}}^{\text{sev}}]$$

where:

$$\mathcal{L}_{\text{CE}}^{\text{cat}} = - \sum_{j=1}^K y_{ij} \log p_{ij}^{\text{cat}}$$

$$\mathcal{L}_{\text{CE}}^{\text{sev}} = - \sum_{j=1}^S y_{ij} \log p_{ij}^{\text{sev}}$$

4.2.3 Hyperparameters

Parameter	Value	Rationale
α	0.7	Prioritize teacher knowledge transfer
τ	4.0	Sufficiently soft distributions for 8 classes
Learning rate	2×10^{-5}	Fine-tuning regime for pre-trained encoder
Steps	5,000	Convergence observed at ~4,000 steps

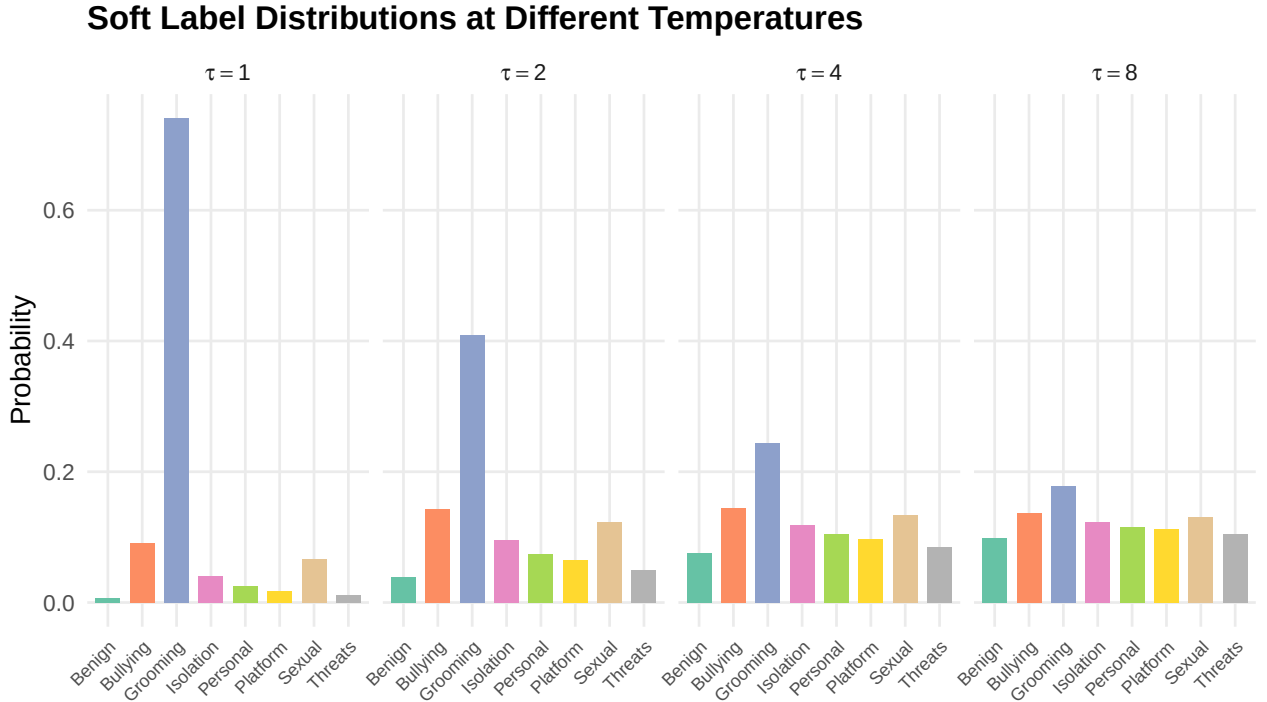


Figure 3: Effect of temperature on soft label distributions. Higher temperatures produce smoother distributions that transfer more inter-class knowledge.

4.3 Gradient of the Distillation Loss

For a single sample, the gradient with respect to student logit z_k^S is:

$$\frac{\partial \mathcal{L}_{\text{KD}}}{\partial z_k^S} = \frac{1}{\tau} (p_k^T - q_k)$$

This shows that the distillation gradient is proportional to the difference between the student’s and teacher’s softened predictions, providing a richer learning signal than one-hot labels alone.

5 Explainability via Integrated Gradients

To provide transparency in risk decisions, we compute token-level attribution scores using Integrated Gradients (IG).

5.1 Method

For an input $x = (x_1, \dots, x_n)$ and a baseline $x' = ([\text{PAD}], \dots, [\text{PAD}])$, the attribution for token i is:

$$\text{IG}_i(x) = (x_i - x'_i) \times \int_{\alpha=0}^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha$$

where F is the model’s output for the predicted class. In practice, we approximate the integral with $m = 50$ Riemann steps:

$$\text{IG}_i(x) \approx (x_i - x'_i) \times \frac{1}{m} \sum_{k=1}^m \frac{\partial F(x' + \frac{k}{m}(x - x'))}{\partial x_i}$$

5.2 Output Format

The top- N tokens (default $N = 5$) by attribution magnitude are returned:

$$\text{attributions} = \text{top}_N \left(\frac{|\text{IG}_i|}{\max_j |\text{IG}_j|} \right)$$

Normalized to $[0, 1]$, these scores indicate which tokens most influenced the classification decision.

6 Deployment Architecture

6.1 Export Pipeline

The mobile model is exported through the following pipeline:

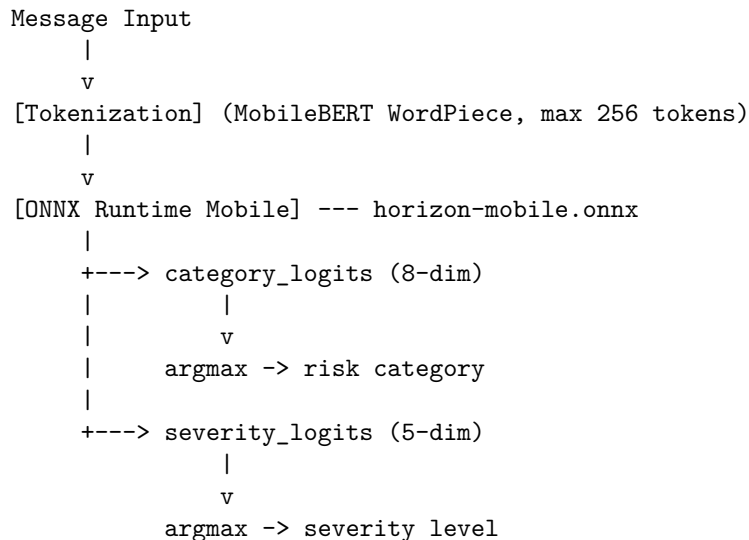
$$\text{PyTorch} \xrightarrow{\text{torch.onnx.export}} \text{ONNX (FP32)} \xrightarrow{\text{quantize_dynamic}} \text{ONNX (INT8)}$$

INT8 dynamic quantization applies the transformation:

$$\hat{w} = \text{round} \left(\frac{w}{s} \right) \cdot s, \quad s = \frac{\max(|w|)}{127}$$

where s is the per-channel scale factor, reducing model size by approximately $4\times$ with minimal accuracy degradation.

6.2 Inference Flow



6.3 Platform Support

Platform	Runtime	Format
iOS	ONNX Runtime Mobile (ObjC/Swift)	horizon-mobile.onnx
Android	ONNX Runtime Mobile (Java/Kotlin)	horizon-mobile.onnx
Server	PyTorch / vLLM	Llama 3.2 3B + LoRA adapter
Desktop (LM Studio)	llama.cpp	GGUF Q4_K_M

7 Training Data

7.1 Synthetic Generation

Training data is generated using Amazon Bedrock (Claude) with category-specific prompts designed to produce realistic conversation patterns. Each conversation undergoes automated quality validation:

- Schema compliance (required fields, types)
- Risk consistency (labels match content)
- Length constraints (minimum conversation turns)
- Diversity checks (no near-duplicate conversations)

7.2 Dataset Statistics

Category	Samples	Proportion
Grooming	8,000	16%
Bullying	7,000	14%
Sexual Content	7,000	14%
Isolation	5,000	10%
Personal Info	5,000	10%
Platform Migration	3,000	6%
Threats	5,000	10%
Benign	10,000	20%
Total	50,000	100%

Training/evaluation split: 90%/10% (45,000 / 5,000 examples).

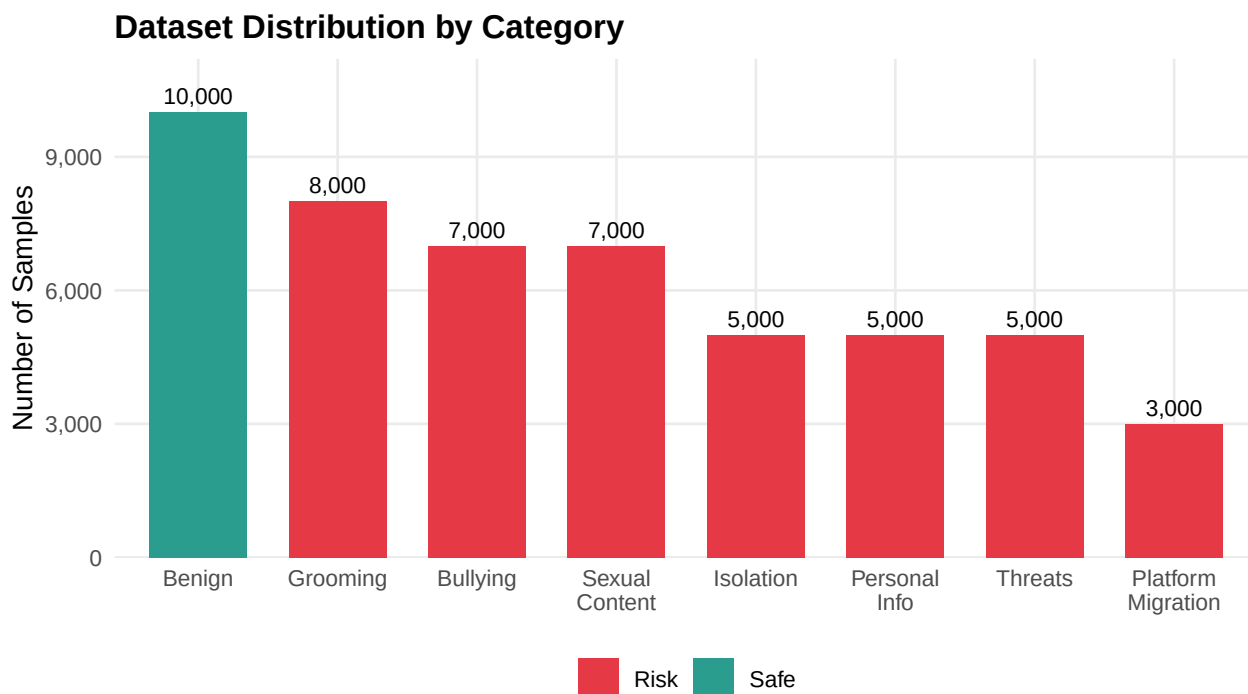


Figure 4: Training dataset distribution by risk category.

7.3 Instruction Format

All training examples follow the Llama 3 chat template:

```
<|begin_of_text|><|start_header_id|>system<|end_header_id|>
{system_prompt}<|eot_id|>
<|start_header_id|>user<|end_header_id|>
Analyze this conversation:
{conversation}<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
{json_output}<|eot_id|>
```

8 Privacy Guarantees

Horizon’s mobile deployment provides the following privacy properties:

1. **Data locality:** All inference occurs on-device. Message content never leaves the user’s phone.
2. **No telemetry:** The model produces classification outputs only; no conversation data is logged or transmitted.
3. **Minimal footprint:** The 25MB model requires no cloud connectivity for inference.
4. **Transparent decisions:** Token-level attributions allow users to understand why a risk was flagged without exposing content to reviewers.

9 Limitations and Future Work

- **500-step checkpoint quality:** Initial results use a quick training run; production deployment requires the full 15,000-step training

- **Language coverage:** Currently English-only; multilingual support requires additional training data
- **Adversarial robustness:** The model has not been tested against adversarial inputs designed to evade detection
- **Temporal context:** Each message is classified independently; multi-turn pattern detection across sessions is not yet supported
- **False positive calibration:** Severity thresholds require tuning against real-world deployment data

10 Conclusion

Horizon demonstrates that effective child safety risk detection can be deployed on-device without compromising user privacy. By combining a high-capacity teacher model with knowledge distillation into a lightweight mobile classifier, we achieve real-time inference with explainable outputs. The system’s architecture enables deployment across iOS and Android without relying on cloud infrastructure, establishing a new paradigm for privacy-preserving content safety.

11 Appendix A: Mathematical Notation Reference

Symbol	Meaning
$\mathcal{C}$	Conversation (sequence of messages)
K	Number of risk categories (8)
S	Number of severity levels (5)
r	LoRA rank
α	LoRA scaling factor / distillation weight (context-dependent)
τ	Temperature for knowledge distillation
W_0	Pre-trained weight matrix
B, A	Low-rank decomposition matrices
η	Learning rate
λ	Weight decay coefficient
$h_{[\text{CLS}]}$	Pooled [CLS] token representation
z^T, z^S	Teacher/student logits
q, p	Teacher/student probability distributions
IG_i	Integrated Gradient attribution for token i

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